

Artificial Intelligence

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Agenda

- Introduction to COMP4431
 - Definition of artificial intelligence
 - Definition of rational agent
 - Assessment
 - Tentative schedule
 - Group project
- History of artificial intelligence
 - Definition
 - Birth
 - Golden time
 - Winter
 - Struggle
 - Revival
- Add/drop consulting
 - Teaching staffs

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What is Artificial Intelligence

- Artificial intelligence
 - Is concerned with the design of intelligence in an artificial device
 - Term coined by McCarthy in 1956
- Artificial device
 - Limited to computer system in our course
- Definition of intelligence
 - Quality that enables an entity to function appropriately and with foresight in its environment
- A typical definition of artificial intelligence
 - A science and a set of computational technologies that are inspired by—but typically operate quite differently from—the ways people use their neuro systems and bodies to sense, learn, reason, and take action
- No widely accepted definition
 - Relationship between machine learning, big data, deep learning, pattern recognition

Rational Agent

- Rational Agent

- An agent is an entity that perceives and acts
- An agent function is to determine actions from percept histories:

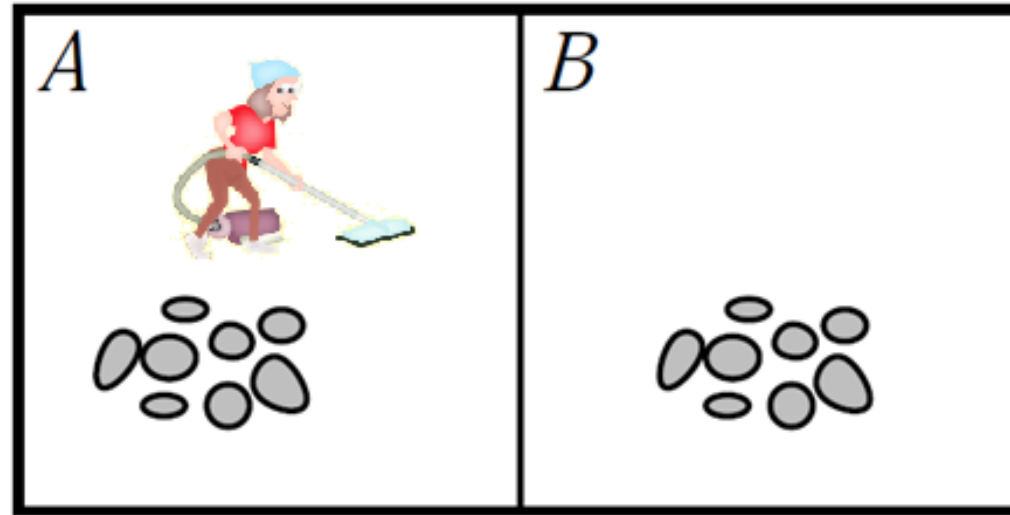
$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

- Characteristics of rational agent

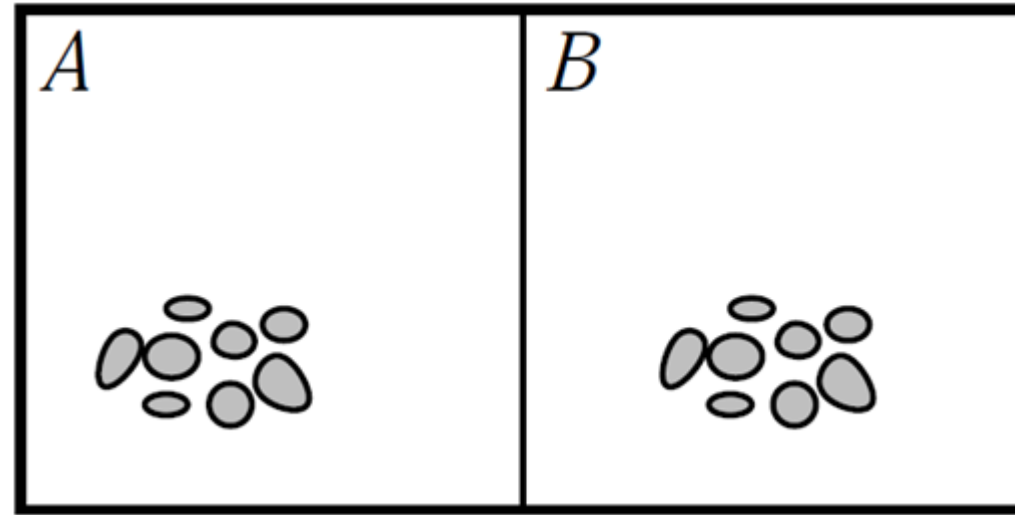
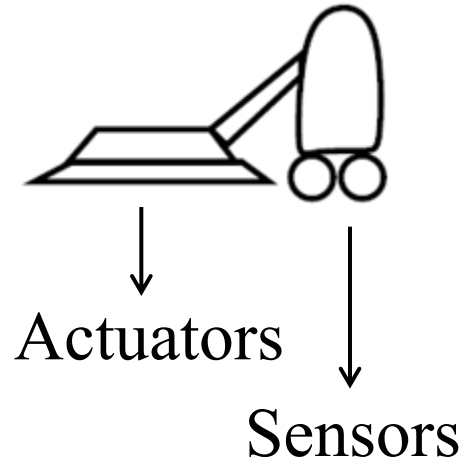
- Distinct from omniscience
- An agent is autonomous if its behavior is determined by its own experience
- Agents can perform actions in order to modify future percepts so as to obtain useful information

A Room Clean Job



- The human perceives the environment
 - which room, clean or dirty
- Decides what to do
 - move right or left, suck the dust
- And then acts

A Vacuum-Cleaner Agent



Environment

- How to perceives the environment
 - which room, clean or dirty
- How to make the decision
 - move right or left, suck the dust
- How to make the action

Subject Learning Outcomes

- Basic knowledge of artificial intelligence
 - Class teaching
 - Quiz and exam
- Basic skill of artificial intelligence implementation
 - Lab teaching
 - Group project
- Basic ability of applied artificial intelligence knowledge and skill in the real problem solving
 - Group project and group presentation

Assessment

■ Individual work

- Quiz 10%
 - Quiz paper will be distributed in the classroom and returned to the TA face to face immediately after the quiz
 - Zero point for late submission
 - If you miss the quiz due to the medical reason and the visa problem, please send email to TA Xingzhao Yan via xingzhao.yan@connect.polyu.hk for recording
 - Makeup quiz is arranged on Tue. April 14
- Final exam 70%
 - Absence and late situation are handled based on the regulations of the university for the examination

■ Group work 20%

- 1-4 members/group
 - The number of members will not influence the grading
- The group members are given same score
 - If not, please send the request to Xingzhao Yan via xingzhao.yan@connect.polyu.hk with the justification and the signature of all group members
- Submission
 - Each group only needs to submit one copy by one group member

Group Project

- Choose one topic from these two candidate ones
 - Group Project I: AI Tutor for AI Learning (15 quotas)
 - Group Project II: AI Professor for AI Teaching (15 quotas)
- Design and implement the Agent
 - Give the presentation to introduce your runnable agent
- AI Tutor for AI Learning
 - Submission deadline: Fri. Mar. 27 23:59
 - 30% deduction/day
 - Presentation date: Tue. Mar. 31 15:30
 - Zero point missing the presentation
- AI Professor for AI Teaching
 - Submission deadline: Fri. Apr. 10 23:59
 - 30% deduction/day
 - Presentation date: Tue. Apr. 14 15:30
 - Zero point missing the presentation

Grading of Group Project

- Design
 - Clear articulation of the problem scope, target users, and core needs (4 points)
 - Innovation & Critical Thinking (2 points)
- Implementation
 - Core proposed features are fully implemented and operational (6 points)
 - The system runs reliably (2 points)
 - System architecture and user interaction design (UI/UX) are logical (2 points)
- Presentation and demo
 - Well –organized presentation (2 points)
 - Answer the question properly (2 points)
 - The requirement of presentation will be announced later

Rubrics of Group Project

- A
 - Have good understanding of the knowledge, and be very good at problem solving and programming
- B
 - Have good understanding of the knowledge, and be very good at problem solving or programming
- C
 - Have good understanding of the knowledge, and good at problem solving or programming
- D
 - Have the basic understanding of the knowledge, and have the basic ability of problem solving or programming
- F
 - Lack the basic understanding of the knowledge or the basic ability of problem solving or programming

Tentative Arrangement

- Lab arrangement will be announced in the lab next week

Week	Date	Topics	Remarks
1	Jan. 13	Introduction to COMP4431	Questionnaire
2	Jan. 20	Heuristic Search	Quiz 1
3	Jan. 27	Knowledge based Agent	Quiz 2
4	Feb. 3	Hill Climbing, Simulated Annealing and GA	Quiz 3
5	Feb. 10	Constraints and Games	Quiz 4
6	Feb. 24	Machine Learning	Quiz 5
7	Mar. 3	Neural Networks	Quiz 6
8	Mar. 10	Deep Learning	Quiz 7
9	Mar. 17	Bayesian	Quiz 8
10	Mar. 24	TBD	Quiz 9
11	Mar. 31	Presentation and Demonstration of AI Tutor	
12	Canceled because of public holiday		
13	Apr. 14	Presentation and Demonstration of AI Professor	Makeup Quiz

Rubrics of Quiz

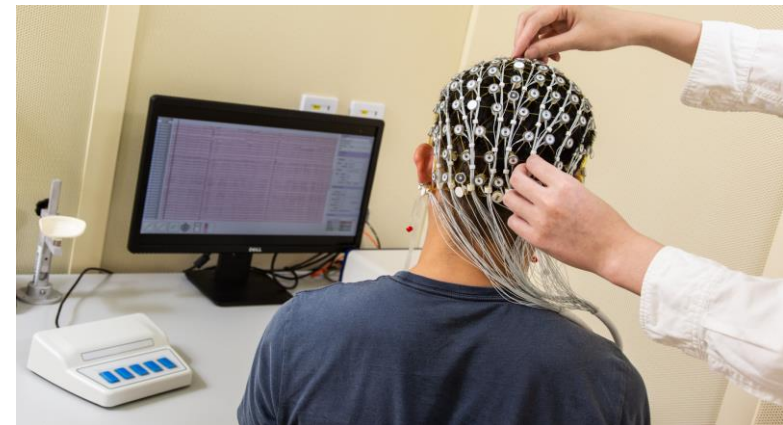
- A
 - Have complete and deep understanding of the knowledge, and be good at problem solving
- B
 - Have complete or deep understanding of the knowledge, and be good at problem solving
- C
 - Have good understanding of the knowledge, and have the basic ability of problem solving
- D
 - Have the basic understanding of the knowledge, and have the basic ability of problem solving
- F
 - Lack the basic understanding of the knowledge or the basic ability of problem solving

General Information

- Text-book
 - ❑ Stuart Russell and Peter Norvig, “Artificial Intelligence A Modern Approach”, third edition.
- Material
 - ❑ Will be available on Blackboard
- Submission
 - ❑ Course project via Blackboard
- Fiona Yan Liu
 - ❑ Responsible for the lecture teaching, grading of the group presentation, grading of the final exam
 - ❑ Research group leader of Artificial Intelligence and Robotics
 - ❑ Artificial intelligence, Brain modeling, Computer music, Affective computing
 - ❑ yan.liu@polyu.edu.hk
- Xingzhuo Yan
 - ❑ Responsible for the lab teaching
 - ❑ Research area: brain imaging
 - ❑ If you have any question, please send email to her by xingzhuo.yan@connect.polyu.hk
- Other TAs
 - ❑ Will be announced later

Advertisement

- Artificial intelligence research group (AIR)
- Recruitment of participants for afternoon nap experiment
 - ❑ 1.5 hours ~ 200 HK\$
 - ❑ PQ610
 - ❑ Everyday 13:00

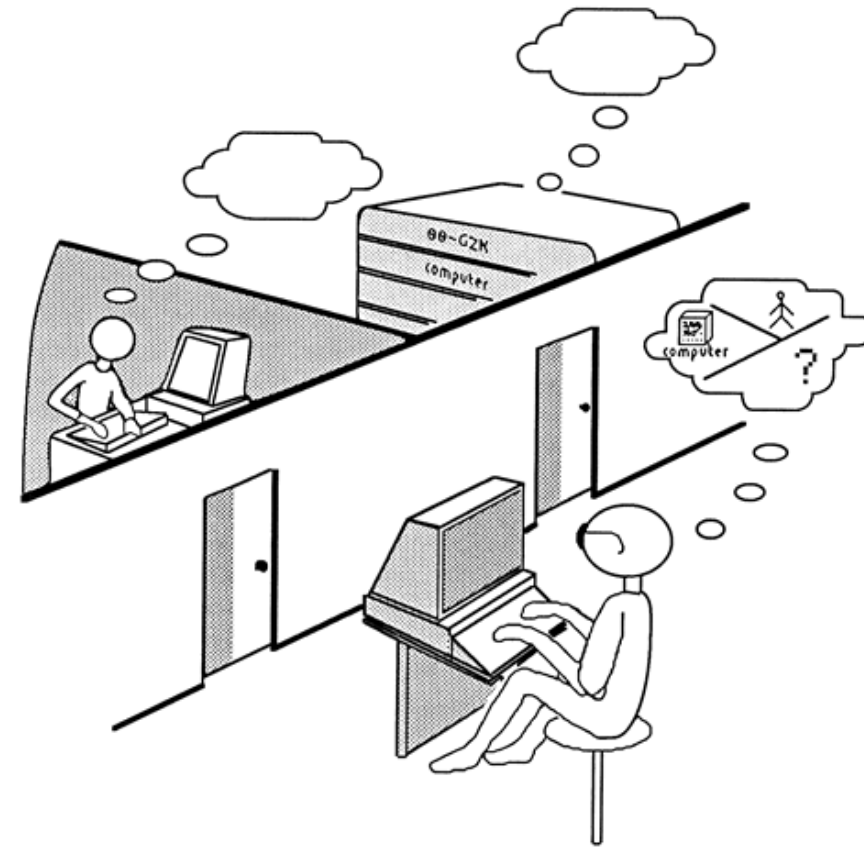


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Turing Test

- Aim to solve the problem
 - ❑ Can machines behave as the human beings
- Proposed in 1950 by Alan Turing
 - ❑ Participate all major argument against AI
- Initial Design
 - ❑ Human
 - Types in questions
 - Receives answers on screen
 - ❑ Computer/human
 - Processes questions
 - Returns answers
 - ❑ Judge
 - Computer or human?
- Limitation
 - ❑ Lack of mathematical analysis
 - ❑ Not reproducible



Alan Turing

- Born
 - In 1912 at United Kingdom
- Died
 - In 1954 (Aged 41)
- Education
 - Undergraduate at King's College in mathematics
 - PhD from Princeton University
- Turing machine
 - Give a definition of computation
 - Absolute limitation on what computation could achieve
- Contribution to artificial intelligence
 - Propose that modification of behavior could be adapted from learning brain to learning machine in 1948
 - Design Turing test in 1950



Dartmouth Colleague

- Dartmouth colleagues
 - Ivy research university
 - Four senior professor retires from the Department of mathematics in 1954
 - John George Kemeny came back to Princeton University for recruitment
 - Inventor of Dartmouth time sharing system
 - John McCarthy has been hired as assistant professor by Dartmouth Colleagues
- Summer in 1955
 - John McCarthy came to work with Nathaniel Rochester
 - IBM 701 computer designer and wrote the first assembler
 - Write the proposal for a summer forum next year
- Team members of proposal
 - John McCarthy, Nathaniel Rochester, Claude Shannon, and Marvin Minsky
 - Shannon is the god father of information theory

Dartmouth Conference



Photo from the 1956 Dartmouth Conference with Marvin Minsky, Ray Solomonoff, Claude Shannon, John McCarthy, Peter Milner, Oliver Selfridge and Nathaniel Rochester.

Five of the attendees of the 1956 Dartmouth Summer Research Project on Artificial Intelligence reunited at the July 2006 AI@50 conference. From left: Trenchard More, John McCarthy, Marvin Minsky, Oliver Selfridge and Ray Solomonoff. (Photo by Joseph Mehling)



John McCarthy

- Born
 - In 1927 at USA
- Died
 - In 2011
- Get Turing Award in 1971
- Professor at Stanford University
- Contribution to artificial intelligence
 - Coined the term of artificial intelligence
 - Organize the Dartmouth conference
 - Created the LISP language in 1958
- Contribution to other areas
 - Came up with a scheme for creating general purpose timesharing



Seven Research Directions of Artificial Intelligence Proposed in 1956

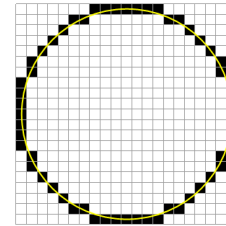
- Automatic computers
 - Autonomous vehicles
- How can a computer be programmed to use a language
 - Python
- Neuron nets
 - Deep learning
- Theory of the size of a calculation
 - GPU
- Self-improvements
 - Machine learning
- Abstractions
 - Feature engineering
- Randomness and creativity
 - Artificial arts

Achievements in the Birth Stage

- 1950 - 1980
- Search
 - Reasoning as search
 - Heuristic search
- Natural language processing
 - Eliza is the first Chatbot created in 1966
 - Eliza has no understanding of the content and use pattern matching to answer the question
- Computer vision
 - Micro-worlds
 - Pixel – shape – object – scene



Pixel



Shape



Object



Scene

Golden Time

- 1980 – 1987
 - The knowledge revolution
- The rise of Symbolicism
 - Symbolic logic is claimed that can explain the character of mind
- Allen Newell and Herbert Simon
 - Proposed logic theorist in 1955
 - Prove the mathematical theorems successfully
- Expert system has been widely used till now
 - Answers questions or solves problems about a specific domain of knowledge
 - Reasoning by logical rules that are derived from the knowledge of experts

Allen Newell and Herbert Simon

- Long-term partners
 - Share the Turing Award in 1975
- Herbert Simon
 - He received the Nobel Prize in Economics in 1978
 - Cognitive psychologist
 - Decision making
- Allen Newell
 - Published the paper of Chess Machine in 1955



Winter

- 1987- 1993
 - The practical progress is far behind the expectation of the public
 - The difficulty to reproduce general intelligence
- Rethinking of the importance of abstract reasoning
 - Building intelligence from bottom to up
 - Trigger the research of perception and interaction with the environment
 - Body is important
 - Enhance the research of robotics
- Embodied artificial intelligence
 - Integrates artificial intelligence into physical entities like robots, making them interact with their physical environment dynamically



Humanoid robot NAO

Struggle

- 1993 – 2011
 - Computer scientists and software engineers avoided the term artificial intelligence for fear of being viewed as wild-eyed dreamers
- The success of deep blue produced by IBM in 1997
 - First winner of computer chess-playing system
- The rise of machine learning
 - With large volume data
- Support vector machines
 - Most powerful binary classifier
- Autonomous vehicles
 - DARPA Grand Challenge



Deep Blue in IBM's headquarters in Armonk, N.Y.



Garry Kasparov vs. Deep Blue

Revival

- 2012 –
- Deep learning
 - The success of connectionism
 - Imitate the physical structure and the information processing of the biological neuro system
- Marvin Minsky
 - Turing Award 1969
 - Co-founder of MIT Artificial Intelligence Lab
- Turing award 2018
 - Geoffrey Hinton
 - Yoshua Bengio
 - Yann LeCun



Generative AI

- Generative artificial intelligence
 - Technique that can be used to create new content, including text, code, speech, music, images, and videos.
 - ChatGPT, Suno, Midjourney, Pika
- ChatGPT
 - Chat Generative pre-trained transformer
 - Developed in 2022 by OpenAI
- Discriminative AI and generative AI
 - Discriminative AI focuses on modeling the relationship between input features and output labels
 - Generative AI focuses on modeling the inherent data structure and generation processes



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